

ActInf Textbook Group ~ Cohort 1 ~ Meeting 12

(Review, Questions, Feedback)

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hey everyone welcome to week of octave textbook group cohort one we are in the review and synthesis weeks it's been a great three-month-ish journey and it doesn't stop here but we'll go over that so two points of process for those who are here live and of course re-watching and then we'll move to the more conceptual parts so first in the onboarding page you'll see these two columns the two columns are rsvping yes to join cohort two part one going through chapters one through five if you would like to for a second time many people will be just doing it for fun or for learning or however they want and also there's the opportunity to continue on with the second half of the textbook which is going to get into a lot of the hands-on modeling and we'll be especially looking forward to developing a lot of the notebooks and interactive formats that'll help us get a lot of understanding there um so please check yes for whichever row your name is in for whether you would like to continue on with part two as well as retake uh part one as well you'd be welcome to be just a participant and also feel free to get in touch if you are interested in taking some other roles such as like facilitating or scaffolding some aspect but we can talk more about that for those who are interested so point one go to the onboarding list and continue on however you want second point um if you go to future textbook groups page you'll see an embedded form as well as a link to the form if you want it in a new window like that this would be exceptionally requested and helpful for you to provide some evaluations numerically and as short or as long as you want on providing feedback those are one way that you can provide feedback anonymously the second way is uh building on some of the ideas that people are already adding here this is an editable page so people can add thoughts as they see fit about like what would be awesome for future textbook groups or any other feedback or as always they can email active inference activeinference@gmail.com if they want like a response to any specific points um and then also it's provided here at the link to share for people who want to join future cohorts so that link will be like getting people onboarded into the next cohort which is september 2022 in this case

but this is going to be like an evergreen form that will just continue to have a list of people who expressed interest and be on boarding them into subsequent cohorts of which will begin several per year going forward any uh and next week we'll be talking more about the feedback specifically and about projects and about carrying on today is like a little bit of a more conceptual synthesis and review and again next week will be more like logistical um and project oriented review so just on these points or anything else does anyone want to just raise their hand or unmute and share anything that they like okay let us conceptually review then um we read the first five chapters of the active inference textbook does anyone have overall thoughts that they'd like to provide at the five chapter scale then we're going to go into a chapter scale then we'll be continuing to dive in as um granularly as required but at the scale of the first five chapters you can see them here on this preface page as well as here in the chapters list as a unit of five chapters what were people's sense how did it update yes please ben and then only well i was just going to say i think um you know as i've said to people before i think coming from a non-mathematics background i thought that these chapters were really accessible in terms of the kind of conceptual uh things that they were putting on the table but i i just wondered what other people's opinions were in terms of you know understanding the conceptual toolkit of active inference before you've built up a kind of mathematical understanding um because i'm about to start kind of um learning some of the mathematics behind it and i wondered what people thought of like the the potential to fully grasp these concepts without the mathematics and what the relationship between the two might be i'm just curious what other people think about that i guess thanks great question um ali and then mike and then anyone else who wants to chime in yeah it was a pretty exciting journey for me uh but i'll definitely need to read over all those five chapters once again at least once again uh in order to grasp uh more fully uh the contents of those chapters uh uh my own personal opinion about uh the way materials uh is organized in these uh the first part of the textbook is uh uh i couldn't see a kind of um hierarchical or uh i don't know structural uh organization for these chapters as much as i i like to see them because um sometimes uh they delve into a theoretical uh theoretical aspect of all the things much more deeply than the mathematical side and sometimes it's vice versa but i i think that if it was organized in a way that as ben mentioned we can go from a firm found a firm conceptual foundation into the granular formalism and mathematics at least for me that would be much more uh accessible and much more uh i would be able to organize the materials in a coherent much more coherent way in my mind so uh one thing uh that um i really enjoyed about uh these chapters is uh the

kind of uh you see uh a kind of uh the vision it uh [Music] tries to put forward uh regarding the future uh possible research uh and also uh things especially in chapter five uh and the the path that can be taken from this point on thank you mike and then anyone else yeah i'll pick up from where ali left off i think it's somewhat remarkable that they were able to fit everything they did into those first five chapters and understanding they probably had some idea of how much space they wanted in that part of the book before getting into the the practical or applied components in part two you know the the high road low road contrast and sort of coming at the problem from two different angles i think was useful like ben i found the math to be generally accessible although i confess i i think i still go through this sort of pattern where i feel like i get it and i have this intuitive understanding of it but then there are things where uh i sort of bog down and looking at some of the mathematical details or some of the graphical representations um related to things like message passing have to slow down and sort of take those apart more and i guess related to that i i should add i think this group has been so good at taking apart the ideas of the book this is perhaps one of the most effective settings i've seen for really ringing concepts out of a text and uh you know the work that's going on with the coda to build up the ontology and take apart the equations and things like that i think is is just tremendous awesome thank you like yeah it's um you know the british call them maths plural instead of math as an area and it really does engage there's visual formalisms which could be represented with a sparsity matrix or it could be represented with other um ways but there are um almost like a multi-scale diversity of formalisms ranging from more traditional equal sign in the middle mathematics to fusion schematic equation graphics different types of notations even within the equation and the difficulty not signaled not that it has to be but the difficulty does provide a little whiplash sometimes because it moves variously in the main thread of the text the boxes and the appendices from like this is how linear algebra works and this is what bayes theorem is all the way to topics that are approaching physics flows on partitioned states various kinds of fundamental or like in principle relationships about mathematics generally postulated architectures where message passing is implemented to achieve some computational function and or thread the needle with resembling neural computational architectures the math is doing a lot many things are happening and and so it's um only to be expected that the perception challenging and there's probably a lot more to even say and um unpack there anyone else want to raise their hands give a thought on this section one area jf and then anyone else have kind of a general question about the limits of mathematics the notion of computational irreducibility that you have to go through every step

of a process to get at state and plus 1 from state and from state and minus whatever certain problems cannot be solved by plugging values into a formula and generating a future state you have to compute all the intermediate states i wonder where that fits in in this picture are there hard limits to the application of of of mathematics um to to to the problem of active inference it's just a general question returning to some um earlier comments on the firm conceptual ground what is an area no matter how narrow or distal from actin where people believe that there is a firm conceptual ground they felt like that provided a foundation for them to learn further are we seeking an analogy to some other field of theory or practice or domain or are we kind of seeking for an epistemic territory that we haven't quite seen i'll just add i think that what in active inference what you're talking about there jeff is literally blankets and uncertainty like the corollary there of like computational irreducibility is that there's some large amount of uncertainty in states in the world that you're just never going to get at the blanket is in some sense not just how you're relating to it but some hard limit on your you know your surprisal and things like that so thanks uh brock had written oh yes mike please uh so responding to the question about maybe adjacent spacex or spaces or topic areas uh i think there's certainly overlap with concepts from system dynamics and certainly a lot of agent-based modeling has been applied to model systems that have feedback loops and adaptation things like that we discussed during the the past weeks relationships with reinforcement learning and i think there's a belief that active inference kind of cleans up some of the challenging or maybe not challenging but uh less formalized or less defined aspects related to reinforcement learning such as uh how do you engineer a utility function that makes sense in the context of what you're trying to model banks agreed about creating a uh integrated utility function there's so many unprincipled not to say ineffective or not even to say not elegant but unprincipled methods of creating a chimera utility model delay discounting novelty bonus curiosity bonuses alternating phases or hyper parameters um as well as purely implementational strategies like discarding burn in parallel chains there's like a whole toolkit and indeed fields on creating effectively integrated utility models putting that work into the construction of the generative model and how the generative model is partitioned from the generative process allows a generic free energy functional or set of related free energy formulations to play that role so that's one very interesting angle the way that that process of specifying the generative model and generative processes and so on are world models whether in the john le sense or in the atom saffron sense these generative models encompass world models and deep learning approaches and various topics like that um so let's see blue wrote i

wish there was practice problems in the book to develop our understanding to a large extent and also it's whether people want to reflect on any specific questions or even just on this scheme these are many of these questions these questions can be open infinite game type questions about the material they can be clearly addressable questions about the material they can be checks for understanding but we'll be continuing to develop and improve and curate questions around the material so i think it's a great comment this is why we have this future textbook groups and all of these affordances for people to stay involved and and be improving this as well as experimenting in their own spaces and ways to to develop the kinds of material that are helping them learn and understand any more section one overview thoughts chapters one through five and the framing of the book being that the first five chapters are more focused on the conceptual background and the second five chapters are going to be starting with a recipe for designing the active models and and getting more into the modeling itself so any section one through five commentary the book comprises two parts page three these are aimed at readers who want to understand active inference first part and those who seek to use it for their own research second part the first part of the book introduces both conceptually and formally contextualizing it within current theories of cognition how did they succeed and where was there a divergence between what you preferred a priori or now comprehensive formal and self-contained introduction to active inference its main constructs and implications for the study of brain and cognition i'm uh i'm interested in the distinction between uh understanding active inference and using it for our own research and the way that that's set up as those being two completely separate things it seems to suggest that one could fully understand active inference and yet not be in a position to use it in their own research not having worked through the second half of the book and i think that's quite interesting it is good comment thank you not to uh spoil the ending but the last sentence is ultimately we are confident that you will continue to pursue active inference in some form there may be a multi-year incubation for different people and contexts many people now or even in some amazing decentralized science future they may not think of themselves as researchers or of doing research though they may be even included on research projects so it's it's very interesting uh comments about how they distinguish like theory and learning from part two research and practice however they discuss applications but this isn't the playbook it's not the toolkit it's not the modeling tool ecosystem mike yeah that's an interesting comment at the end i know this is my perceptual bias but there seems to be this implicit assumption that active inference is the way to go basically this is the tool to use within the text so as a result what's not there is a

discussion of uh maybe these are cases where you might not want to use active inference or these are cases where applying active inference could create challenges in what you're solving for ali and anyone else and in fact one of the uh most popular criticisms put toward the active inference or fep in general is that it tries to explain everything it ends up explaining nothing yeah i think there's a risk of maybe not sufficiently constraining the problem space or leaving things undefined in a way that allow for active inference to be the solution without a critical point of view highlight behavior and cognition i mean one could almost see even the name active inference as being a synthesis action behavior and cognition inference in ways that as we're discovering and unpacking they're integrated in ways that other formalisms of perception as a type of cognition so we can just lump it there other frameworks of behavior and cognition have not integrated or have approached from a non-first principles or maybe they do follow first principles like the higher the impact factor the better the paper something like that that's the first principle no just kidding um placing it within the framework of while also surfing on a wave something like pan cognitivism or like pan it computationalism expand the scope and that's even before one starts to enter into the more recent especially where there is a highlight on system persistence not simply in terms of the resistance to dissipation entropically but in a relational context as repeated measurements like in the quantum work it's like cognition oh like brains like that schema with a brain or with like a person well oh by cognition it's any system with a blanket that we're interacting with through repeated measurement is that cognition where did cognition go brock i am i kind of want to i guess echo all those things that were just said about the common criticism and this where does it apply where does it not apply but also i guess um all those things that you were just listing like i i wonder i always hear that from people that have not um engaged with the material and from my engagement with it it seems like all of the examples in which it is used it are complex dynamical systems that don't have great that have at best mathematical approximations and have almost no or literally no kind of analytical approaches it just seems like a good way to model a system of things of entities that are interacting and i don't i i guess what i'm to to put it into like one question like what is uh is that kind of some anthropomorphizing or whatever projecting of like well it's applied to this it's applied to that it must apply to everything and um you know how can it apply to all those things how can it apply everywhere it's like it's not applying to everywhere it's just applying to these complex systems which are everywhere but also don't have and you know we're not trying to model the apple falling from the tree or the hyper you know the these sorts of simple cases or something it's not being applied there i

mean is it i don't i haven't seen actually you know now that i think about it any of those sorts of situations or it's applied to things that we already understand in this like maybe in some back-checking way but not in a like way that's suggested that we should use this much more complex you know approach to something that's kind of check already got that thanks brock um ali i kind of feel that uh some of these criticisms uh stems from some from the fact from a serious a lack of deep understanding of uh concepts related to fap and active for example uh i came across a phd thesis just today in which it tries to critique everything related to mark of blanket as i was glancing over it uh i observed some serious uh gaps uh in our argument throughout all and all the whole the thesis and i think uh materials such as this book uh can definitely help in filling up these gaps of understanding which even the the serious researchers are suffering from yes thanks so to um this thread i've never heard it about the linear how could the linear model be used in so many settings it explains nothing well is the linear model an explanation or is it an investigative tool in the investigators toolbox at least for me personally doesn't need to have broader reach than those who resonate with it but again i always try to ask would somebody say this or could they say that about a linear model and then if the actin formalism f equals dot or g equals dot and all of the predicates that it's associated with and the partition and all of that could someone say that about y equals mx plus b and then if not and there may be cases where it is not the same what is it that's different about active or is it a valid and important and even reasonable question but one that might be bumping up like what jf raised about computational or about map territory even disguised or camouflaged questions about relationality well how can complexity science apply to so many topics we're applying complexity science to whatever topic we want to where's the issue mike and then ron yeah a lot of it hinges on how the model is to be applied and what the intent of model application is so taking the linear model as an example we might use something like that to be predictive about a linear system and so there's potential value there uh when we do system dynamics models a lot of times the value in doing a system dynamics model is aligning the stakeholders around what are the model components and how do we think about those model components so working through a process similar to like what we've done in this textbook group although few do it as rigorously identify the system elements and how they interact and interrelate and so forth and and that can lead to more advanced implementation like agent-based models but a lot of times at least in my experience those agent-based models are used for deeper systems understanding more than say predictive power for what the system is going to do at some point in the future so again it comes back to the the reason for applying

the models and thinking about how the model fits with that reason and what sort of results you would expect from that thank you mike ron uh yeah i i generally agree with what you and mike said so from a practitioner's perspective i mean we do hear criticisms of linear models right it's not that we don't but it's usually about which domain is being applied in yeah uh but generally speaking well uh let me close this with uh because active inference is probably more complex and it takes a lot more time to drop than $y = mx + b$ which could be one reason for these kind of criticisms and it's probably like that they're falling back into old habits okay why should i jump my old as opposed to adopting this newer more well how much more does it explain than what i hope i think that probably is driving a lot of the criticism of these things i can't help but add a domain specific example um this is a 2016 paper that is speaking to like a multi-decade multi-career brouhaha different approaches to modeling evolution and selection for example in the eu social insects and in social animals as well and this article is very fascinating they use a bayesian causal graph to identify situations where two different formulations the kin and the multi-level selection models have identical predictions where the causal graphs have identical predictions any measurement you can think of as being like on a $y = mx + b$ manifold where it doesn't resolve your uncertainty about which framework is correct or not yet the literature is littered with we measured these ovaries so can selection is or isn't happening however a vast set of those empirical cases might be essentially falling on this manifold where those two models are not distinguishable and so this is a theory driven approach like a first principles approach to identify situations providing unique informative value in this case about canon multi-level selection and knowing what territories measurements will not have explanatory value which relates to active inference um just like rohan and like many people have brought up especially when we kind of take like a meta science or like a communicating science or onboarding people into active concern like people are asking explicitly or implicitly why should i update my cognitive model what is the value of active inference can you appetize me with a two minute video so i can understand it or with a two minute video so it's enough to want to continue going down that path there's a lot to say and there's um a lot of work to identify the situations and the ways of truly addressing and perhaps even resolving long-standing scientific divergences if it is the case that the explore exploit dialectic is addressed in a novel way through the free energy functional balancing pragmatic and epistemic reward that is quite a vast scope people continue to use explore and exploit today and it is evident that those ontology terms wouldn't be useful in the future in fact some of the discussions on like folk psychology with the belief desires intentions

can we say what an active inference entity wants or intends based upon its beliefs and its desires for example can we still talk about exploratory and exploitative behavior as a phenomena but have a different way of modeling how that behavior arises or or is regulated what are the settings in behavioral cognitive science where systems dynamics has been applied where we can now use active inference to identify where have we been just throwing a thousand darts at the same grain of sand where are the vast fields where we don't even know and as people have highlighted the book is written with a specific rhetorical bent both explicitly and implicitly for the regime of attention of the book to be about active inference one can read as an implicit endorsements that this is something that is valid or valuable to pursue especially at this stage in the field it is almost like an exhortation persist amidst uncertainty to learn and apply few disjointed thoughts there but i hope it makes sense let's look back to the subjections of the book anyone can raise their hand or give a thought like could the book you write begin what reviewer comments whether you're reviewer one reviewer two reviewer three what comments would you have provided if this were like a draft and you could even make structural suggestions so i have a lot of comments there yes blue and then ron um i uh so like having taught lots of classes before like i think um a list of like key terms and definitions would be like critical i mean i mean and learning objectives for each chapter like i said uh like a practice quiz like test your knowledge of this like like what is surprise like how many different forms of surprise are there you know i mean like like just some practice questions to see if you grasped the concepts prevented in each chapter and especially like uh what i was mentioning in the chat like math problems like practice like can you set up your own generative model um like like what would the like or even just the bayesian graph like can you make a bayesian graph like what would that look like for you know give a give a question like a verbal description and then give the bayesian graph like answer or like make the answers available online so you could test your own knowledge of the subject first um i think that that would be like totally instrumental in in using this functionally like a textbook anyway so sorry thanks blue nice predictive programming rohan and then brock uh yeah uh so my as a practitioner right i work in the engineering field i'm a control systems engineer my preference would be to immediately see some application not necessarily to control systems but something that it begins with like a project and explains how active inference is better than its own baseline that would be interesting to see uh like a perfect example of this would i don't know if you've heard of fastai by rachel thomas and jeremy howard they have a very unique way of teaching so they have a computational linear algebra course so the way they teach the

linear algebra part is by basically implementing gaussian blur for example and then connecting it to so you start with the code and then you do the math then you connect it to the math so something like that might be far more useful uh maybe not from a point but it would definitely be useful for practitioners around the world okay so this so if you know controls engineer gets this okay this looks like a better form model predictive control i can replace my model predictive controller with something that resembles active inference it might actually make uh we might actually make more progress that it actually gets out of the way thank you ron brock yeah i i think i mean i've had similar comments to that of um fast ai and benchmarks that that's kind of the ultimate way that's just the way that um practitioners change their um attention now um and that would be really useful but in in the context of the book i think i agree with every everything else that was said too uh i'm not sure how realistic it is to do that from the point you know where the whole field is at right now sorry about that i think just having some some things in context like there's always this comment about you know it's a non-linear thing a text is linear so of course it'll be out of out of order but like um often the definitions of things are kind of scattered throughout the explanation of them italicize instead of kind of brought up to the top you know in some way you might expect that i at least from my pedagogical experience everything like all math is taught to you know like to define your knowns and unknowns and variables at the top and the sort of thing right um the in the context of like the appendices again there's like i understand that it's a lot you can't you know do you can't add that all in or whatever but just if there was like um either every whenever the next part of the appendice became relevant you know um again just like having it at the top like at each section you're like okay so we're gonna use part you know now building on you know what we've done already like now we're going to use this part of this this section in the appendices in this section our equation one two three whatever in in in this next section like just stating it up front again just like a like a you know big box on the top or whatever it just would help like in wayfinding and disentangling unfortunate linear um format of books that is mapped to this complex thing um yeah i don't know those those general things would be helpful one other thing that is maybe beyond the scope of this book but um spoke to ellie about very briefly at one point about the notation like even in the basic example of the frog here um like bayesian notation it's incredibly easy to manipulate variables when they're one letter and subscripts but when the whole variable is like 10 or 20 letters like you know frog p p of x equals frog jump you know condition on a y it's like it becomes a larger i don't i don't know if this is just me and dyslexic you know like i don't know but um it just seems harder to cognitively kind of um

work with that i don't know thank you brock yes the chunking is multi-level and there it will be great to explore ways to um learn around that here are a few forays where we can copy quotes and have inline definitions so for those who are interested in this work with an open source textbook it is not an infinite task or even a non-automatable task to copy out the plain text and integrate it with a versioning ontology so that the definition can be like even rendered differently it could be in a different language it could be every single time that a term comes up there's like so much enrichment that this seed can lead to of the textbook a lot of it is for us to pick up honestly rohan and then anyone else oh i i forgot to put down oh no worries thank you so in our um last seven or so minutes for this session next week we will continue with any conceptual points and questions that people want to raise we'll also be turning our attention a bit more uh making sure first off that please please everyone complete the feedback in the future textbook groups and or add information here or um contact us it's just one of the most important feedback mechanisms we have we'll also be exploring a little bit different project ideas several of these threads involving different system models we've already found a home for in the active block prints project where we're meeting twice on wednesdays weekly other projects we're going to have new vistas on during and after going through the second half of the is uh still black and white static pdfs but we'll be able to augment this heavily with some simulation tools work like the plain text enrichment or even anyone who wants to collaborate on an audiobook version there's um no lack of act imp tasks today and these are like high leverage point moments where staying in the game and improving the game will do tremendous service towards reducing research debt and improving the rigor and the applications that we see rohan yeah uh can i make a project suggestion that's fine yes please everyone can yeah so that yeah okay so there is uh i'm gonna put this in the chat there's this online lab by georgia tech's intelligence assistance unit uh called robutarioro that's basically a bunch of swarm it's to test swarm controllers so uh there's a good baseline over here it would be a good idea to try like an active inference version like an active inference inspired swarm controller to finish the tasks over here so this runs on actual robots in that lab and you can see the results this is awesome um i had some colleagues who worked with some of the ant groups here it's right at the intersection of jf's work and many's interest in embodied and robotic angle and we have somewhat extensive multi-scale generative models of ants which would apply to other swarm settings as well so absolutely we want to scaffold those projects and one way to start but not the only way to start is to copy a version of this these fields and add and remove as relevant for you and give some handle for others to be involved or just stay in the game so

that um more people can join there will be a flow of people entering active inference forever how will we greet them so uh what would i need to put in it depends on how you want to carry forward the project but you can feel in this row add any notes you'd like more details on what you're thinking of doing and then however you want to be more specific about like directions just signal it in the idea or you can in the um notes or or draft catechism which is what these templates are based on and if there is ever something you're sure about then ask around and um different projects are going to proceed differently okay yeah uh so i'll fill that out with my talk so awesome well interesting meeting it's our penultimate discussion in this section then we hope that everybody who is motivated and ready to do so continues for part two and or rejoins for part one cohort two both of these will be starting in september so we'll take august um off of the textbook uh at this point the tentative plan is that we'll have two sequential hours section one and then followed by section two so that those who are on section one can um see how section two is we can explore different architectures and stuff and again this is like why people's active participation and their feedback and the over sharing on topic and just writing all the questions that came in their mind and all the things that they would want to ask somebody to check like every contribution is important does anyone else have any thing they'd like to add before we stop the recording stopping the recording oh yeah someone go for it all right stopping the recording